

Experimental Plan

Experiment	0 min	10 min	20 min	30 min	40 min	50 min	60 min	Objective / Test Note
1	0	1	1	1	0	0	0	Baseline Settling: 1 person stays 30 mins, guaranteeing the 15-min transient period completes and the EKF reaches a stable steady-state estimate.
2	0	1	1	2	2	0	0	Staggered Steady-State: 1 person settles, then a 2nd enters and stays 20 mins. Proves the filter can cleanly transition and settle between discrete steps.
3	1	1	2	1	1	0	0	Short-Visit Disturbance: 1 person is already settled. 1 friend drops in for <10 mins and leaves before the 15-min room settling time finishes.
4	2	2	2	4	2	2	0	High-Density Transient: 2 people settled. 2 friends drop in for a rapid <10 min visit. Tests if the EKF can handle an extreme CO ₂ generation spike without lagging.
5	0	2	2	3	3	3	0	Sustained High Load: 2 people settle, then 1 friend joins and stays for 30 mins. Proves the observer accurately settles at the 3-person boundary.

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6	0	1	0	2	2	2	0	Rapid Swap & Settle: 1 person makes a quick visit (fails to settle), followed immediately by 2 people who hold for 30 mins to force a clean steady-state recovery.